NIHVA Technologies Pvt. Ltd.

Annexure A

Scope

Transferring of Coated Layers from the Applicator machines to the feed zone of the ELPO curing oven.

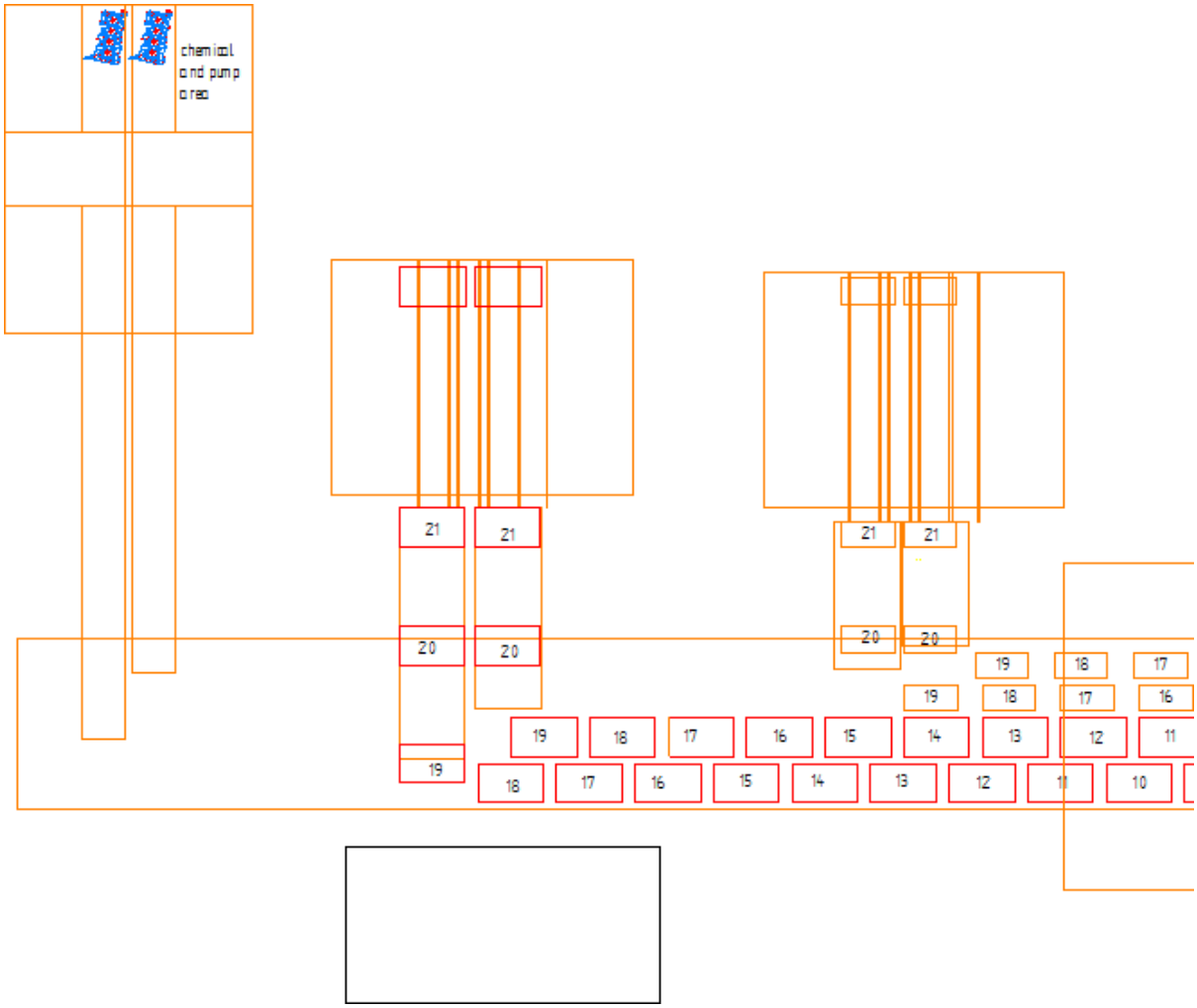
General Information

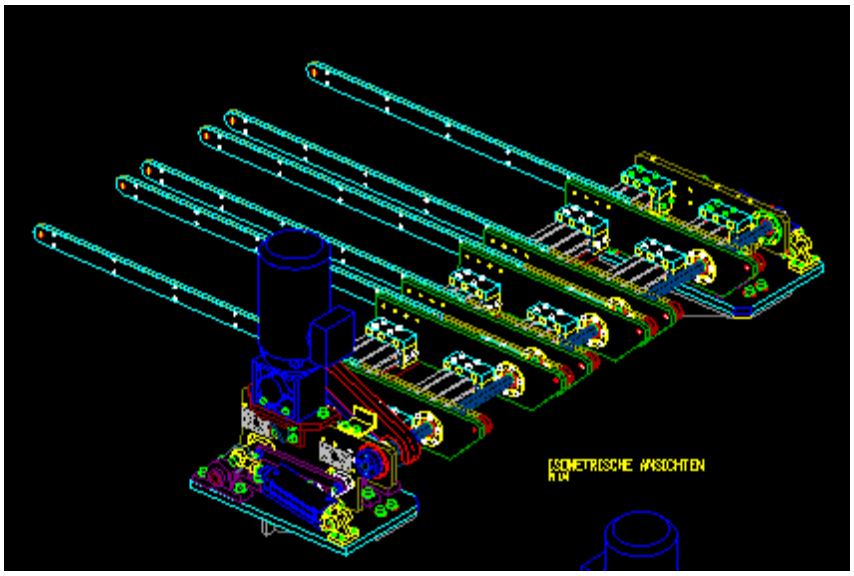
Screen Printing Machine (SPT)

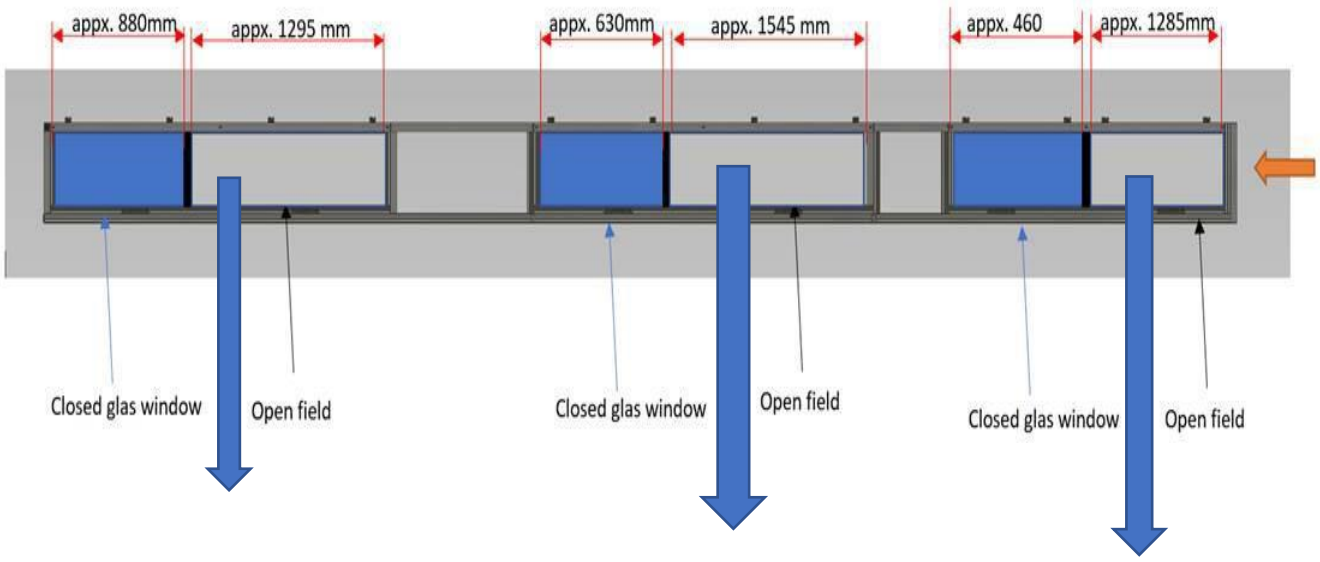
- We have 2 Screen Printed machines (SPT) with double cavities. Each cavity has output rate of one layer at a time, i.e. 2 layers output per stroke of the screen print machine. Each cavity has 2 belts on which layers are pushed out of the screen print machine. We plan to have 4 belts for 2 screen printed machines. (this means we have 1 belt on each cavity, so for 4 cavities we have 4 belts)
- Each belt needs will have different length for dropping the layers in the Feed zone, so the belt length should be adjustable. Please see the diagram below.

Roller Coating Machine (NBR)

- We have 1 Roller Coating Machine (NBR). We have a normal conveyor belt for feeding the layers. 2 layers are fed per stroke of the machine. We plan to have 2 belts for NBR.
- Belt length should be adjustable. Please see the diagram below.


chemical
and pump
area
21
21
20
20
19
21
21
20
20
19
18
17
19
18
17
16
19
18
17
16
15
14
13
12
11
18
17
16
15
14
13
12
11
10


ISOMETRISCHE ANSICHTEN



Belt 5-6 Belt 4-3 Belt 1-2

•The Feed zone has a Cover on it with the above dimension.

- There are 3 openings on one side and one opening on the other side for SPT 1 machine.
- the opening is as follow:

The opening is as follow:

For NBR: 1285 mm
For SPT 1: 1545 mm
For SPT 2: 1295 mm

- Height of Each Opening is 400 mm maximum

Dimension of the layers to be transferred from SPT and NBR are as follow:

	Min	Max
Length	100mm	600mm
Width	100 mm	400mm
Height	0.1 mm	1.0 mm
Weight	14 gram	200 gram

Process flow

- The operator scans the bar code from the production order paper at the main curing oven HMI panel. The signal goes at first to the oven main panel.
- In the main panel there are predefined program which will be provided by oven supplier according to OT requirements. As the signal goes to main oven panel the curing oven start to reach the required parameters.
- Once the parameter has reached the tolerance the oven will send a signal to the transfer mechanism (which belts to run), accordingly the belt or belts will start running.
- The operator can now start his production on the screen print machine.

2. Transfer Mechanism

Mechanical Specification

Belts	Belt 1	Belt 2	Belt 3	Belt 4	Belt 5	Belt 6
	NBR		SPT-1		SPT-2	
Length(mm) (Minimum)	1100	1100	750	750	650	650
Length(mm) (Maximum)	1700	1700	1700	1700	1300	1300
Length(mm) (Fixed)	1000	1000	650	650	650	650
Width (mm)	330	330	650	650	650	650
Height (mm) (Adjustable) (from ground to the top of the belt)	950-1100	950-1100	950-1100	950-1100	950-1100	950-1100
Belt Material	PVC	PVC	PVC	PVC	PVC	PVC
Belt Speed (m/min)	0-10	0-10	0-10	0-10	0-10	0-10
Idler Diameter for the belt (mm)	50	50	50	50	50	50

Technical Specifications:

- Belt Conveyor:
  - Makes no imprints or scratches on the layers
  - Belt material should be PVC
  - Belt durability
  - Easy to change
  - Belt tension should be maintained all the time
  - Belt aligning must be provided
- Motor should be servomotor. NOTE: The motor will be near explosive area. (Suitable option for the same can be used with prior approval from OT)
- Gap between belts
  - Belt 1 & Belt 2 – 40 mm
  - Belt 3 & Belt 4 – 40 mm
  - Belt 5 & Belt 6 – 40 mm
- The whole Transfer Mechanism system will rest on floor with resting pads and wheels are also need to be provided for easy movement.
- The conveyor belts should be open. (Belt case covering)
- Conveyor orientation should be flat throughout the length.
- The construction needs to be built in such a way that no big movements will be happen during using.
- To ensure the correct positioning each unit could be fix with a physically structure with the feed zone
- No vibration has to be there at free end of all conveyors. If required support can be given from other side with prior approval from EK)
- Aluminium profiles to be used for structure.
- There should be no deflection in angle of the layers while transferring from machine end to Feed zone belt. If required additional structure can be added in feed zone to achieve zero deflection.

Electrical Parameters

- The Transfer mechanism will be connected with the main oven line for communication between Transfer mechanism and Oven
- The Connection Interface will be PROFInet
- PLC Connection type
  - Siemens 7-1515F-2PN
- PN/PN coupler coupling module for connecting two PROFInet network

- Model No. 6ES7158-3ADO1-OXAO
- Mounting: mounting rail 7.5mm and 15mm
- Quantity -1

- HMI Panel
- One HMI Panel and emergency stop button for all the Belts

3. Additional line information

- Spare part package (optional)
- All local and country specification must be followed
- Supplier logo is defend to put on the machine
- Oven colour RAL 7035 (light grey)
- Quote should have DAP price and ex works price.
- Final length version will be decided during the project.

4. Energy

- According to the specific local rules and OT specification D
- Power connection (water air pressure & electric) according to the OT specification D
- Power connection should be in one area.
- All local rules and country specification must be followed (India)
- In case of using an electrical cabinet an UPS backup has to be installed cause of India power failure (power bank again power failure to avoid electrical damage)

5. Training

- Maintenance Training required
- Worker training required

6. Documentation

- Country safety standards local laws must be followed
- Complete documentation including user manual, electrical circuit diagram & function diagrams, floor plans, Wear & Spare Parts List: 1x paper, 2xCD-ROM.
- Complete Documentation according to the OT specification D
- Maintenance plan & maintenance contract according to OT order
- Information about the surface stress and the footprint of the machine and a breakdown of dynamic and static load calculation / components

7. Acceptance

- Acceptance contents must be discussed between supplier and OT.
- Pre-acceptance must be made with parts by the supplier
- During the Pre-acceptance OT and the supplier inscribe all the fault and defect on the machine. (important for the final approval)
- Between Pre-acceptance and final acceptance, the supplier must solve all the defects and fault which both has inscribe by the Pre-acceptance.
- Between Pre-acceptance and final acceptance, the supplier must solve all the defects and fault which both has inscribe by the Pre-acceptance.